Percentage of dissociation $= \sqrt{1.7 \times 10^{-4} \times 100}$

$= 1.3 \times 10^{-2} = 0.013\ \%$

[Option (b)]

11. $pH = -\log_{10}[H^+]$

$\therefore [H^+] = 10^{-pH}$

Let the volume be x mL

$V_1M_1 + V_2M_2 + V_3M_3 = VM$

$\therefore$ x mL of 10^{-1}M + x mL of 10^{-2}M + x mL of 10^{-3} M

= 3x mL of $[H^+]$

$$\therefore [H^+] = \frac{x[0.1+0.01+0.001]}{3x}$$

$$= \frac{0.1+0.01+0.001}{3}$$

$$= \frac{0.111}{3}$$

$= 0.037$

$= 3.7 \times 10^{-2}$

[Option (a)]

12. $AgCl_{(s)} \rightleftharpoons Ag^+(aq) + Cl^-(aq)$

$\underset{0.1\ M}{NaCl} \longrightarrow \underset{0.1\ M}{Na^+} + \underset{0.1\ M}{Cl^-}$

$K_{sp} = 1.6 \times 10^{-10}$

$K_{sp} = [Ag^+][Cl^-]$

$K_{sp} = (s)(s+0.1)$

$0.1 >>> s$

$\therefore s + 0.1 \simeq 0.1$

$$\therefore S = \frac{1.6 \times 10^{-10}}{0.1} = 1.6 \times 10^{-9}$$

[Option (b)]

13. $PbI_2(s) \rightleftharpoons Pb^{2+}(aq) + 2I^-(aq)$

$K_{sp} = (s)(2s)^2$

$3.2 \times 10^{-8} = 4s^3$

$$s = \left(\frac{3.2 \times 10^{-8}}{4}\right)^{1/3}$$

$= (8 \times 10^{-9})^{1/3}$

$= 2 \times 10^{-3}$M [Option (a)]

14. Addition of salt KY (having a common ion Y^-) decreases the solubility of MY and NY_3 due to common ion effect.

Option (a) and (b) are wrong.

For salt MY, $MY \rightleftharpoons M^+ + Y^-$

$K_{sp} = (s)(s)$

$6.2 \times 10^{-13} = s^2$

$\therefore s = \sqrt{6.2 \times 10^{-13}} \simeq 10^{-7}$

for salt NY_3,

$NY_3 \rightleftharpoons N^{3+} + 3Y^-$

$K_{sp} = (s)(3s)^3$

$K_{sp} = 27s^4$

$$s = \left(\frac{6.2 \times 10^{-13}}{27}\right)^{1/4}$$

$s. \simeq 10^{-4}$

The molar solubility of MY in water is less than of NY_3

[Option (d)]

15. x ml of 0.1 M NaOH + x ml of 0.01 M HCl

No of moles of NaOH = $0.1 \times x \times 10^{-3} = 0.1x \times 10^{-3}$

No of moles of HCl = $0.01 \times x \times 10^{-3} = 0.01x \times 10^{-3}$

No of moles of NaOH after mixing = $0.1x \times 10^{-3} - 0.01x \times 10^{-3}$

$= 0.09x \times 10^{-3}$

$$\text{Concentration of NaOH} = \frac{0.09x \times 10^{-3}}{2x \times 10^{-3}} = 0.045$$

$[OH^-] = 0.045$

$P^{OH} = -\log(4.5 \times 10^{-2})$

$= 2 - \log 4.5$

$= 2 - 0.65 = 1.35$

$pH = 14 - 1.35 = 12.65$

[Option (d)]

Answers.indd 299 11-01-2021 17:36:14